Topics Covered:

- Exponential Representation of Group Motion / Displacements for $SE(3)$ and $se(3)$
- Exponential Representation of Group Motion / Displacements for $SE(2)$ and $se(2)$
- Notes on Rotations in $SE(3)$

Additional Reading:

- MLS Chapter 2, Section 3.3; LP 3.3.3

Recall from last class that we discussed the exponential representation of group motion for rotations only ($SO(3)$ and $\mathfrak{so}(3)$). This allows us to compute $R \in SO(3)$ using the Lie-algebra element ($[\omega]_\times \in \mathfrak{so}(3)$) using Rodrigues' formula:

$$e^{[\omega]_\times \tau} = I + \frac{[\omega]_\times}{\|\omega\|}\sin(\|\omega\|\tau) + \frac{[\omega]_\times^2}{\|\omega\|^2}(1 - \cos(\|\omega\|\tau)),$$

We also discussed how this formula can be equivalently represented as a rotation about an axis $\hat{\omega}$ through an angle $\theta = \|\omega\|\tau$:

$$e^{[\hat{\omega}]_\times \theta} = I + [\hat{\omega}]_\times \sin(\theta) + [\hat{\omega}]_\times^2(1 - \cos(\theta)),$$

Alternatively, we can compute the logarithm which solves for an axis and angle ($\hat{\omega} \in \mathbb{R}^3$ and $\theta \in \mathbb{R}$) given $R \in SO(3)$:

If $\|\hat{\omega}\| = 1$, then $(\hat{\omega}, \theta) = \ln R$ is given by:

$$\theta = \cos^{-1}\left(\frac{\mathrm{Tr}(R) - 1}{2}\right) \begin{cases} \theta = 0, & \hat{\omega} \text{ is arbitrary} \\ 0 < \theta < \pi, & \hat{\omega} = \frac{1}{2\sin(\theta)}\begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \\ \theta = \pi, & \hat{\omega} = \pm\frac{c_k}{\|c_k\|}, \quad c_k = \text{any nonzero column of } R + I \end{cases}$$

If $\|\omega\| \neq 1$, then $\omega = \ln(R)$ is given by:

$$\|\omega\| = \cos^{-1}\left(\frac{\mathrm{Tr}(R) - 1}{2}\right)$$

$$\hat{\omega} = \frac{\omega}{\|\omega\|} = \frac{1}{2\sin(\|\omega\|)}\begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

with the same special cases: if $\|\omega\| = 0$, then $\omega = 0$, and if $\|\omega\| = \pi$, the axis $\hat{\omega}$ is found from a nonzero column of $R + I$ as above.

We will now extend this to $SE(3)$ and $se(3)$, and then to $SE(2)$ and $se(2)$. This process will look similar except we will obtain:

$$g = \underbrace{e^{\hat{\xi}\tau}}_{\text{general}} = \underbrace{e^{\hat{\xi}\theta}}_{\|\omega\|=1} \qquad \text{(The exponential map)}$$

$$\underbrace{(\xi, \tau)}_{\text{general}} = \underbrace{(\xi, \theta)}_{\|\omega\|=1} = \ln g \qquad \text{(The logarithm)}$$

Note that ξ is not the same twist in the two forms: in the θ form it is scaled so that $\|\omega\| = 1$, and θ is the angle rotated (so $\hat{\xi}\tau$ and $\hat{\xi}\theta$ are equal as products).

Exponential Representation of Group Motion / Displacements

II. $SE(3)$ and $se(3)$

For this case $\xi = \begin{bmatrix} v \\ \omega \end{bmatrix}$ and $\hat{\xi} = \begin{bmatrix} [\omega]_\times & v \\ 0 & 0 \end{bmatrix}$

We won't derive, just will give the answer:

a) if $\omega = 0$, then

$$\hat{\xi} = \begin{bmatrix} 0 & v \\ 0 & 0 \end{bmatrix} \implies \hat{\xi}^n = 0 \ \text{ for } n \geq 2$$

Therefore,

$$e^{\hat{\xi}\tau} = \begin{bmatrix} I & v\tau \\ 0 & 1 \end{bmatrix}$$

b) if $\omega \neq 0$, then

$$e^{\hat{\xi}\tau} = \begin{bmatrix} e^{[\omega]_\times \tau} & (I - e^{[\omega]_\times \tau})\frac{[\omega]_\times}{\|\omega\|^2} v + \frac{\omega\omega^\top}{\|\omega\|^2} v\tau \\ 0 & 1 \end{bmatrix}$$

Alternative form for translation:

$$
\begin{aligned}
&(I - e^{[\omega]_\times \tau})\frac{[\omega]_\times}{\|\omega\|^2}v + \frac{\omega\omega^\top}{\|\omega\|^2}v\tau \\
&= \left[I - \left(I + \frac{[\omega]_\times}{\|\omega\|}\sin(\|\omega\|\tau) + \frac{[\omega]_\times^2}{\|\omega\|^2}(1-\cos(\|\omega\|\tau))\right)\right]\frac{[\omega]_\times}{\|\omega\|^2}v + \frac{\omega\omega^\top}{\|\omega\|^2}v\tau \\
&= -\left(\frac{[\omega]_\times^2}{\|\omega\|^3}\sin(\|\omega\|\tau) + \frac{[\omega]_\times^3}{\|\omega\|^4}(1-\cos(\|\omega\|\tau))\right)v + \frac{\omega}{\|\omega\|}\frac{\omega^\top}{\|\omega\|}v\tau \\
&= \frac{\|\omega\|^2[\omega]_\times}{\|\omega\|^4}(1-\cos(\|\omega\|\tau))v - \frac{[\omega]_\times^2}{\|\omega\|^3}\sin(\|\omega\|\tau)v + \left(I + \frac{[\omega]_\times^2}{\|\omega\|^2}\right)v\tau \\
&= I\tau v + \frac{[\omega]_\times}{\|\omega\|^2}(1-\cos(\|\omega\|\tau))v + \frac{[\omega]_\times^2}{\|\omega\|^3}(\|\omega\|\tau - \sin(\|\omega\|\tau))v \\
&= \left(I\tau + \frac{[\omega]_\times}{\|\omega\|^2}(1-\cos(\|\omega\|\tau)) + \frac{[\omega]_\times^2}{\|\omega\|^3}(\|\omega\|\tau - \sin(\|\omega\|\tau))\right)v
\end{aligned}
$$

The Logarithm: given $g = \begin{bmatrix} R & d \\ 0 & 1 \end{bmatrix} \in SE(3)$, what is $(\xi, \tau) = \ln g$?

Recall that $\xi = \begin{bmatrix} v \\ \omega \end{bmatrix}$, so need to figure out for v and for ω.

1. Find the axis and angle $(\hat{\omega}, \theta) = \ln R$, then choose a time τ and set $\omega = \hat{\omega}\theta/\tau$. (Choosing $\tau = 1$ gives $\omega = \hat{\omega}\theta$, while choosing $\tau = \theta$ gives $\omega = \hat{\omega}$.)

2. use formula of $g = \exp(\hat{\xi}\tau)$ to solve for v.

$$
\begin{aligned}
\vec{d} &= (I - R)\frac{[\omega]_\times}{\|\omega\|^2}v + \frac{\omega\omega^\top}{\|\omega\|^2}v\tau \\
&= \left((I-R)\frac{[\omega]_\times}{\|\omega\|^2} + \frac{\omega\omega^\top}{\|\omega\|^2}\tau\right)v
\end{aligned}
$$

$\Longrightarrow$

$$
\begin{aligned}
v &= \left((I-R)\frac{[\omega]_\times}{\|\omega\|^2} + \frac{\omega\omega^\top}{\|\omega\|^2}\tau\right)^{-1}\vec{d} \\
&= \|\omega\|^2\left((I-R)[\omega]_\times + \omega\omega^\top\tau\right)^{-1}\vec{d}
\end{aligned}
$$

or if we use the other method

$$
v = \left(I\tau + \frac{[\omega]_\times}{\|\omega\|^2}(1-\cos(\|\omega\|\tau)) + \frac{[\omega]_\times^2}{\|\omega\|^3}(\|\omega\|\tau - \sin(\|\omega\|\tau))\right)^{-1}\vec{d}
$$

3. What if $\theta = 0$ (so $R = I$ and $\omega = 0$)? Then no rotation, but there may still be translation.

$$
\underbrace{v = d,\ \tau = 1}_{\text{unit time}} \quad \text{or} \quad \underbrace{\tau = \|d\|,\ v = \frac{d}{\|d\|}}_{\text{unit velocity}}
$$

Example 1 (logarithm): Find the unit axis $\hat{\omega}$, angle θ, and twist $\xi = (v, \hat{\omega})$ such that $g = e^{\hat{\xi}\theta}$ for

$$g = \left[\begin{array}{c|c} R_x(\pi/4) & \begin{matrix}2\\3\\0\end{matrix} \\ \hline 0 & 1 \end{array}\right] = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 3 \\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Step 1, the angle and axis from R:

$$\theta = \cos^{-1}\left(\frac{(1+\sqrt{2})-1}{2}\right) = \frac{\pi}{4}, \qquad \hat{\omega} = \frac{1}{2\sin(\pi/4)}\begin{bmatrix} \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \\ 0 - 0 \\ 0 - 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Step 2, solve for v using the formula above with $\omega = \hat{\omega}$ (so $\|\hat{\omega}\| = 1$) and $\tau = \theta$:

$$v = \left((I - R)[\hat{\omega}]_\times + \hat{\omega}\hat{\omega}^\top\theta\right)^{-1} d$$

$$(I - R)[\hat{\omega}]_\times + \hat{\omega}\hat{\omega}^\top\theta = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 - \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & -\frac{\sqrt{2}}{2} & 1 - \frac{\sqrt{2}}{2} \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} \frac{\pi}{4} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\pi}{4} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{2} & -(1 - \frac{\sqrt{2}}{2}) \\ 0 & 1 - \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix}$$

This matrix is block diagonal, so the x component is $v^1 = 2/(\pi/4) = 8/\pi$, and the lower 2×2 block (with determinant $2 - \sqrt{2}$) gives:

$$\begin{bmatrix} v^2 \\ v^3 \end{bmatrix} = \frac{1}{2-\sqrt{2}}\begin{bmatrix} \frac{\sqrt{2}}{2} & 1 - \frac{\sqrt{2}}{2} \\ -(1 - \frac{\sqrt{2}}{2}) & \frac{\sqrt{2}}{2} \end{bmatrix}\begin{bmatrix} 3 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{3}{2}(1+\sqrt{2}) \\ -\frac{3}{2} \end{bmatrix}$$

Therefore,

$$\xi = \begin{bmatrix} v \\ \hat{\omega} \end{bmatrix} = \begin{bmatrix} 8/\pi \\ \frac{3}{2}(1+\sqrt{2}) \\ -3/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \approx \begin{bmatrix} 2.546 \\ 3.621 \\ -1.5 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \qquad \theta = \frac{\pi}{4}$$

As a check, the slide along the axis is $\hat{\omega}\hat{\omega}^\top v\theta = (8/\pi)(\pi/4) = 2$, which is exactly the x component of d.

From an angle to a speed and time: Because $\hat{\omega}$ is a unit vector, θ above is an angle, not a time. Suppose we instead want the motion to take a desired time T. Only the product $\hat{\xi}\theta$ matters, so we can scale the twist by θ/T and apply it for a time T:

$$g = e^{\hat{\xi}\theta} = e^{(\frac{\theta}{T}\hat{\xi})T} = e^{\hat{\xi}_T T}, \qquad \xi_T = \frac{\theta}{T}\xi$$

Now $\xi_T = (v_T, \omega)$ is a velocity: $\omega = \frac{\theta}{T}\hat{\omega}$ is the angular velocity, and $v_T = \frac{\theta}{T}v$. For example, if we want the motion to take $T = 2$ seconds, then $\theta/T = \pi/8$ and

$$\omega = \frac{\pi}{8}\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}\pi/8\\0\\0\end{bmatrix}, \qquad v_T = \frac{\pi}{8}\begin{bmatrix}8/\pi\\ \frac{3}{2}(1+\sqrt{2})\\ -3/2\end{bmatrix} = \begin{bmatrix}1\\ \frac{3\pi}{16}(1+\sqrt{2})\\ -\frac{3\pi}{16}\end{bmatrix}$$

Choosing $T = 1$ is the same as folding the angle into the twist, $\xi_T = \theta\xi$, which is what MATLAB's `logm` returns.

Example 2 (exponential): Now let's go the other way, and check that applying ξ_T from above for $t = T = 2$ seconds brings us back to g. Here $\|\omega\| = \pi/8$ is not 1, so we use the general formulas, and the body rotates through the angle $\|\omega\|t = \pi/4$.

First, the rotation from Rodrigues' formula. Since $[\omega]_\times/\|\omega\| = [\hat{\omega}]_\times$:

$$e^{[\omega]_\times t} = I + \sin(\tfrac{\pi}{4})[\hat{\omega}]_\times + (1-\cos(\tfrac{\pi}{4}))[\hat{\omega}]_\times^2 = \begin{bmatrix}1 & 0 & 0\\ 0 & \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2}\\ 0 & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\end{bmatrix} = R_x(\pi/4)$$

Then, the translation $(I - e^{[\omega]_\times t})\frac{[\omega]_\times v_T}{\|\omega\|^2} + \frac{\omega\omega^\top}{\|\omega\|^2}v_T t$, one piece at a time:

$$\frac{[\omega]_\times v_T}{\|\omega\|^2} = \frac{\omega \times v_T}{\|\omega\|^2} = \frac{64}{\pi^2}\cdot\frac{\pi}{8}\begin{bmatrix}0\\ \frac{3\pi}{16}\\ \frac{3\pi}{16}(1+\sqrt{2})\end{bmatrix} = \begin{bmatrix}0\\ \frac{3}{2}\\ \frac{3}{2}(1+\sqrt{2})\end{bmatrix}$$

$$(I - e^{[\omega]_\times t})\frac{[\omega]_\times v_T}{\|\omega\|^2} = \begin{bmatrix}0 & 0 & 0\\ 0 & 1-\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2}\\ 0 & -\frac{\sqrt{2}}{2} & 1-\frac{\sqrt{2}}{2}\end{bmatrix}\begin{bmatrix}0\\ \frac{3}{2}\\ \frac{3}{2}(1+\sqrt{2})\end{bmatrix}$$

$$= \frac{3}{2}\begin{bmatrix}0\\ (1-\frac{\sqrt{2}}{2}) + \frac{\sqrt{2}}{2}(1+\sqrt{2})\\ -\frac{\sqrt{2}}{2} + (1-\frac{\sqrt{2}}{2})(1+\sqrt{2})\end{bmatrix} = \begin{bmatrix}0\\3\\0\end{bmatrix}$$

$$\frac{\omega\omega^\top}{\|\omega\|^2}v_T t = \hat{\omega}\hat{\omega}^\top v_T t = \begin{bmatrix}1\\0\\0\end{bmatrix}(1)(2) = \begin{bmatrix}2\\0\\0\end{bmatrix}$$

Adding the two translation pieces gives $d = (2, 3, 0)$, so

$$e^{\hat{\xi}_T t} = \left[\begin{array}{c|c} R_x(\pi/4) & \begin{matrix}2\\3\\0\end{matrix} \\ \hline 0 & 1\end{array}\right] = g.\checkmark$$

Theorem (Chasles)**.** *Every rigid body motion can be realized by a rotation about an axis combined with a translation parallel to the axis.*

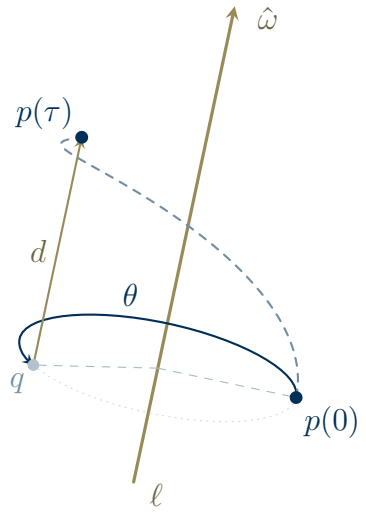


Figure 1: Chasles' theorem. Whatever path the body takes from $p(0)$ to $p(\tau)$, the net displacement can always be reproduced in two steps about a single line ℓ: rotate through $\theta = \|\omega\|\tau$ about ℓ (navy), carrying $p(0)$ to q, then translate by d *parallel to that same axis* (gold). Because the slide is along the very axis of the turn, the two steps commute, and performing them together traces the dashed screw path. The axis direction is $\hat{\omega}$; in general ℓ does not pass through the origin.

In other words, the exponential of a twist represents the *relative* motion of a rigid body:

$$p(\theta) = e^{\hat{\xi}\theta} p(0)$$
$$g_{ab}(\theta) = e^{\hat{\xi}\theta} g_{ab}(0)$$

Here ξ is a **spatial** twist, so the exponential multiplies on the **left**. A **body** twist multiplies on the **right**:

$$g_{ab}(\theta) = e^{\hat{\xi}_s\theta} g_{ab}(0) = g_{ab}(0)\, e^{\hat{\xi}_b\theta}, \qquad \hat{\xi}_s = g_{ab}(0)\, \hat{\xi}_b\, g_{ab}(0)^{-1}$$

Equivalently, $\hat{\xi}_b\theta = \ln\left(g_{ab}(0)^{-1} g_{ab}(\theta)\right)$ and $\hat{\xi}_s\theta = \ln\left(g_{ab}(\theta)\, g_{ab}(0)^{-1}\right)$. Using the wrong side (e.g., $e^{\hat{\xi}_b\theta} g_{ab}(0)$) does not reproduce $g_{ab}(\theta)$. The point equation $p(\theta) = e^{\hat{\xi}\theta} p(0)$ likewise uses the spatial twist.

Theorem (Euler)**.** *Any orientation $R \in SO(3)$ is equivalent to a rotation about a fixed axis $\hat{\omega} \in \mathbb{R}^3$ through an angle $\theta \in [0, 2\pi)$*

III. $SE(2)$ and $se(2)$

exp : for $\xi \in \mathbb{R}^3$ where $\hat{\xi} \in se(2)$,

$$e^{\hat{\xi}\tau} = \begin{cases} \text{if } \omega = 0, & \begin{bmatrix} I & v\tau \\ 0 & 1 \end{bmatrix} \\ \text{if } \omega \neq 0, & \begin{bmatrix} e^{[\omega]_\times \tau} & \frac{1}{\omega}(I - e^{[\omega]_\times \tau})\mathbb{J}v \\ 0 & 1 \end{bmatrix} \end{cases}$$

where we will use the variable $\mathbb{J}$ to represent the matrix $\mathbb{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

ln: given $g \in SE(2)$,

$$\theta = \text{atan2}(R_{21}, R_{11}), \qquad \omega = \theta/\tau$$

for a chosen time τ. If $\theta = 0$ (so $\omega = 0$),

$$v = \vec{d},\ \tau = 1 \quad \text{or} \quad v = \frac{\vec{d}}{\|d\|},\ \tau = \|d\|$$

else,

$$\frac{1}{\omega}(I - R)\mathbb{J}v = \vec{d}$$

$\Longrightarrow$

$$\begin{aligned} v &= \omega \left[(I - R)\mathbb{J}\right]^{-1} \vec{d} \\ &= \omega \mathbb{J}^{-1}(I - R)^{-1}\vec{d} \\ &= -\omega \mathbb{J}(I - R)^{-1}\vec{d} \end{aligned}$$

These expressions for planar motion are summarized as:

$$\exp(\hat{\xi}\tau) = \begin{cases} \left[\begin{array}{c|c} R(\xi^3\,\tau) & \frac{1}{\xi^3}(I - R(\xi^3\tau))\mathbb{J}\begin{bmatrix}\xi^1\\ \xi^2\end{bmatrix} \\ \hline 0 & 1 \end{array}\right] & \text{if } \xi^3 \neq 0 \\ \left[\begin{array}{c|c} I & \begin{bmatrix}\xi^1\\ \xi^2\end{bmatrix}\tau \\ \hline 0 & 1 \end{array}\right] & \text{if } \xi^3 = 0, \end{cases}$$

$$\ln_\tau(g) = \begin{cases} \omega = \frac{1}{\tau}\text{atan2}(R_{21}, R_{11}) & \\ v = -\omega\mathbb{J}\left(I - R\right)^{-1} d & \text{if } \omega \neq 0\,, \\ v = \frac{1}{\tau}d & \text{if } \omega = 0 \end{cases} \quad \text{with } \xi = \begin{bmatrix} v \\ \omega \end{bmatrix},$$

$$\mathbb{J} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Other Representations of Rotation in $SO(3)$

Exponential coordinates are called the *canonical* coordinates of the rotation group. Other coordinates are Euler angles and Quaternions.

Euler angles

When working in $SO(3)$, there are three elementary rotations about the $x-$, $y-$, and $z-$axes:

$$R_x(\phi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix},$$
$$R_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix},$$
$$R_z(\psi) = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

One common method of utilizing Euler angles to describe the orientation of some coordinate frame B relative to another coordinate frame A is to use the ZYZ convention. This means that the orientation of frame B relative to frame A is described by a triple of angles (α, β, γ). These angles describe three rotations: first the rotation of B about the z-axis of B by α, then a rotation about the new y-axis of B by β, and finally a rotation about the new z-axis of B by γ:

$$R_{ab} = R_z(\alpha)R_y(\beta)R_z(\gamma)$$
$$R_{ba} = R_z(-\gamma)R_y(-\beta)R_z(-\alpha)$$

Different ordered sets of rotation axes are also possible. Common choices are ZYX (Fick angles) and YZX axes (Helmholtz angles). The ZYX Euler angles are also referred to as the yaw, pitch, roll angles:

$$R_{ab} = R_z(\psi)R_y(\theta)R_x(\phi).$$

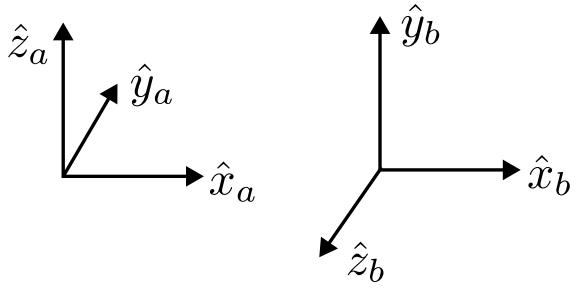
One advantage of these other orders (ZYX and YZX) is that the singularity does *not* occur at the identity orientation ($R = I$). For ZYX Euler angles, the singularity occurs when $\theta = \pm\frac{\pi}{2}$.

In general, singularities occurs when one degree of freedom is lost and the system becomes "gimbal locked" - essentially, losing the ability to independently control one axis of rotation;

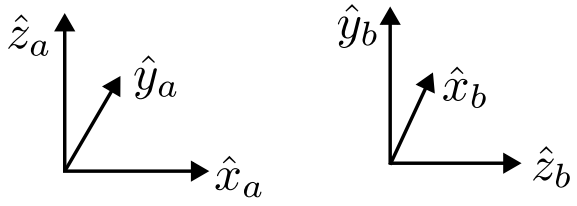
Euler angles are an example of a *local* parameterization of SO(3). It is a fundamental topological fact that singularities can never be eliminated in any 3-dimensional representation of SO(3).

Example

Let's consider the following examples:



$$\begin{aligned} R_{ab} &= R_x(\pi/2) \\ &= R_z(0)R_y(0)R_x(\pi/2) \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$



$$\begin{aligned} R_{ab} &= R_x(\pi/2)R_y(\pi/2) \\ &= R_z(\pi/2)R_y(0)R_x(\pi/2) \\ &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

Quaternions

Unlike Euler angles, quaternions are a *global* representation of SO(3), at the cost of four numbers instead of three. Overall, quaternions are a four-dimensional extension of complex numbers, and can be used to represent rotations in a similar way that complex numbers on the unit circle represent planar rotations. The vector form of a quaternion is:

$$Q = q_0 + q_1 i + q_2 j + q_3 k,$$

where q_0 is the scalar component, and $\vec{q} = (q_1, q_2, q_3)$ is the vector component.

The inverse of a quaternion is given by:

$$\begin{aligned} Q^{-1} &= Q^*/\|Q\|^2, \\ Q^* &= (q_0, -\vec{q}), \\ \|Q\|^2 &= Q \cdot Q^* = q_0^2 + q_1^2 + q_2^2 + q_3^2. \end{aligned}$$

Quaternion products are given by:

$$Q \cdot P = (q_0 p_0 - \vec{q} \cdot \vec{p}, q_0\vec{p} + p_0\vec{q} + \vec{q} \times \vec{p}).$$

Unit quaternions are quaternions with a norm of 1 ($\|Q\| = 1$). Notably, unit quaternions form a group with respect to quaternion multiplication. This allows us to do the following.

Given a rotation matrix $R = \exp([\hat{\omega}]_\times \theta)$, the corresponding unit quaternion is:

$$Q = (\cos(\theta/2), \hat{\omega}\sin(\theta/2))$$

Alternatively, given a unit quaternion $Q = (q_0, \vec{q})$, the corresponding rotation matrix can be computed as:

$$\begin{aligned} R &= \exp([\hat{\omega}]_\times \theta), \\ \theta &= 2\cos^{-1}(q_0), \\ \hat{\omega} &= \begin{cases} \frac{\vec{q}}{\sin(\theta/2)} & \text{if } \theta \neq 0, \\ \text{arbitrary} & \text{otherwise.} \end{cases} \end{aligned}$$

For more information on 3D rotations and quaternions, I recommend the following youtube videos: https://www.youtube.com/watch?v=zjMuIxRvygQ&t=165s
https://www.youtube.com/watch?v=d4EgbgTm0Bg